



PUBLIC-DATA SCENARIO DISCOVERY

CityShift

A clear view of what makes a city's driving environment different - and which scenarios may be most valuable to test first.

CityShift uses public road, infrastructure, weather, and crash data to show how a city differs from selected Waymo benchmark environments and to recommend the driving scenarios that may be most valuable to test first.

CITY COMPARISON

SCENARIO PRIORITIZATION

RIDER CONFIDENCE

Project overview | September 2026

What CityShift does

Every city creates a different driving environment. CityShift turns those differences into a structured, explainable starting point for operations, simulation, testing, and rider-facing safety research.

Find the difference

Measures roads, controls, activity, infrastructure, and climate using the same method in every city.

Locate it

Places every result on a consistent H3 hex map so teams can see exactly where a condition occurs.

Explain it

Shows the three strongest reasons behind each area's Shift score instead of returning a black-box label.

Turn it into action

Builds ranked candidate scenarios that can guide simulation and real-world test planning.

The core question

What is different about driving here, where does it happen, and what should be tested first?

The project does not claim that Waymo lacks these scenarios or that public data can replace internal fleet knowledge. It is a complementary discovery and prioritization layer.

How a city analysis works

1	Search The user selects a city through place search.
2	Divide An 8 km radius is divided into H3 resolution-8 hexagons, about 0.74 km ² each.
3	Measure Public datasets are converted into consistent road, infrastructure, activity, and climate metrics.
4	Compare Each kept hex is compared with pooled areas from five benchmark cities.
5	Prioritize Unusual conditions become ranked candidate scenarios with map locations and evidence.

Benchmarks, hexes, and data

The same geometry and calculations are used for every analyzed city. This makes the comparison repeatable and keeps the result tied to visible evidence.

Selected Waymo benchmark environments

Benchmark city	Reference hexes	Role
Phoenix, Arizona	229	Warm, low-rain road environment
San Francisco, California	161	Dense urban road environment
Los Angeles, California	245	Large arterial and motorway network
Austin, Texas	232	Fast-growing mixed urban environment
Atlanta, Georgia	286	Wettest benchmark climate in the current model
Total pooled reference	1,153	Shared baseline for road and place metrics

These are project-selected benchmark environments, not an official statement of Waymo's current operating footprint.

What each hex measures

Road network Intersection density Road density	Road mix Arterial share Motorway share One-way share	Traffic control Signal density Crosswalk density
Mobility Bike-lane density Transit-stop density	Local activity School density Nightlife density Tourism density	Infrastructure Bridges Movable bridges Tunnels Roundabouts Stadiums

Additional context includes average lane count, terrain slope, driving side, map-data completeness, and the number of usable hexes. Terrain slope and average lanes do not currently affect the Shift score.

Public data sources

- **OpenStreetMap**: road network, intersections, crossings, signals, mobility features, places, and infrastructure.
- **Open-Meteo**: average annual rain, heavy-rain, and snow days for 2020-2024.
- **Copernicus DEM GLO-90**: elevation context delivered through Open-Meteo.
- **NHTSA FARS**: US fatal-crash locations and context for 2020-2024; 2024 data is preliminary.
- **Google Maps and Places**: city search, map display, and location context.

Shift scores and scenarios

A Shift score describes difference, not danger. It shows how unusual one hex is compared with the benchmark distribution and explains the measurements behind that result.

Score	Map label	Meaning
Below 80	Within baseline	Similar to the benchmark distribution
80 to 94.9	Notable shift	Meaningfully different and worth reviewing
95 or higher	Strong shift	More unusual than at least 95% of reference areas

Explainable by design

Each hex lists its three strongest differences, including the local value, benchmark median, statistical distance, and benchmark percentile. Rare features, such as movable bridges in the current reference, are marked as novel instead of being forced into an unreliable score.

Candidate scenario families

Weather and traction	Heavy rain at dense intersections; reduced traction and hidden lane markings during snow.
Vulnerable road users	Pedestrians on wide arterials; cyclists at complex intersections; late-night pedestrian activity.
Road operations	Bus pull-outs, opening movable bridges, multi-lane roundabouts, and tunnel transitions.
Demand surges	Stadium crowds, rideshare pickup congestion, and nearby lane changes.
Traffic conventions	Left-hand traffic and mirrored turn logic relative to the right-driving benchmarks.

How priority is calculated

Scenario priority increases with both the strength of the difference and the number of affected hexes. This helps separate isolated curiosities from repeatable conditions that may deserve broader testing.

What the product delivers

Interactive map Color-coded hexes and exact affected locations.	Why panel The evidence behind a selected hex.	Comparison view City metrics beside the benchmark.
Scenario list Ranked candidates tied to map areas.	Crash context US fatal-crash points shown separately.	Briefing and chat A report plus evidence-grounded exploration.

Helping Waymo enter a new city

Starting operations in a new city requires more than copying a test plan from an existing market. CityShift can provide an early, public-data view of local conditions before large amounts of fleet data are available.

1. Early screening	Run the same analysis across candidate cities to compare the scale and type of local differences.
2. Readiness research	Identify weather, infrastructure, traffic conventions, and activity patterns that need special attention.
3. Simulation planning	Translate unusual combinations into concrete scenario candidates before physical deployment.
4. Mapping and collection	Use the highlighted hexes to focus high-definition mapping, observation, and data-collection routes.
5. Closed-course and on-road tests	Select representative locations and conditions rather than searching the entire city manually.
6. Ongoing regression	Save locally important scenarios and rerun them after perception, planning, or policy updates.

Where the value comes from

A smaller search space

Teams can start with the areas and conditions that differ most instead of treating an entire city as equally unfamiliar.

Locally relevant tests

A city's own roads, weather, activity, and infrastructure shape the proposed scenarios.

Evidence before scale

Public data provides a useful first pass before extensive internal mapping and fleet observations exist.

A shared language

Maps, metrics, and scenario cards give operations, simulation, safety, and product teams a common reference.

Example

A city with heavy rain and dense crossings

CityShift may find that several downtown hexes combine more rain than any benchmark city with unusually dense intersections and crossings. That evidence can guide simulation of reduced visibility, wet braking, signal interpretation, and pedestrian behavior - then point teams to representative locations for validation.

CityShift does not make the launch decision. It helps teams ask better questions earlier, document why certain tests were selected, and use limited testing time more deliberately.

Safe Journey for older adults

Older adults may feel more comfortable when a ride is predictable, clearly explained, and avoids historically concerning corridors when a reasonable alternative exists. Safe Journey is designed to make that tradeoff visible without promising that any route is risk-free.

The idea

Most navigation systems optimize mainly for travel time. Safe Journey also considers historical fatal-crash exposure along the road network and can offer fastest, balanced, and lower-exposure route options.

How it can feel safer and more comforting

Choice without complexity

A rider can choose among a small number of clearly named options instead of interpreting a dense technical map.

Visible tradeoffs

The interface explains the extra time and estimated exposure reduction, making a calmer route feel intentional rather than mysterious.

Fewer concerning corridors

When possible, the system can prefer routes farther from clusters of historical fatal crashes.

Predictable guidance

Clear turn instructions, route labels, and consistent visual language can reduce uncertainty during the ride.

Caregiver confidence

A family member or caregiver can understand why a route was recommended and what tradeoff was made.

Respect for rider preference

Some riders value arriving a few minutes later if the journey feels less stressful and avoids historically severe locations.

A simple rider flow

Choose pickup and destination

Compare 2-3 route options

Select the preferred comfort tradeoff

Follow clear guidance

Important safety boundary

Historical crash exposure is not a prediction that a crash will occur. A lower-exposure route is not guaranteed to be safer, and the feature should never replace current road conditions, emergency guidance, accessibility needs, or Waymo's own safety systems. The benefit is greater transparency and rider comfort, not a safety guarantee.

Where CityShift fits

CityShift is most valuable as an early research and prioritization layer: it helps teams decide where to look, what to test, and how to explain the choice.

New-market screening Compare candidate cities using one consistent process.	Simulation-suite creation Turn local differences into scenario candidates.	Regression testing Preserve locally important scenarios across software updates.
On-road test planning Find representative areas for specific conditions.	Safety-review preparation Provide traceable evidence for prioritization decisions.	Data-quality review Reveal where incomplete public data requires validation.

Important limitations

- A high Shift score means different, not automatically dangerous.
- A low score does not prove that an area is safe.
- Public map data may be incomplete or outdated.
- Weather is city-level rather than hex-level.
- Fatal-crash information is US-only and is not normalized by population or traffic volume.
- Terrain slope is collected but currently excluded from scoring.
- The analyzed area covers an 8 km radius around the selected center.
- Candidate scenarios require expert validation before safety decisions or deployment.
- The project does not replace Waymo's internal maps, telemetry, simulation, validation, or operational judgment.

Project position

CityShift is a decision-support tool

It combines public evidence into a repeatable view of local driving differences. Its purpose is to narrow the search space, reveal candidate scenarios, support new-city preparation, and make route choices easier to understand - while leaving final safety and operational decisions to validated Waymo systems and experts.

One-sentence description

CityShift uses public road, infrastructure, weather, and crash data to show how a city differs from selected Waymo benchmark environments and to recommend the driving scenarios that may be most valuable to test first.

It can support early planning for new-city operations and provide clearer, more reassuring route choices for older adults through the Safe Journey experience.